United States Patent	12410074
Kind Code	B1
Date of Patent	September 09, 2025
Inventor(s)	Coccellato; Cannon

Rain water collection, filtration, and life boat system

Abstract

The invention provides a rain water collection, filtration, and life boat system. The system comprises a rain catcher unit with a life boat having a water tank, and a connecting hose. The connecting hose connects the rain catcher to the water tank of the lifeboat to allow a flow of collected rain water from the rain catcher to the water tank. The connecting hose comprises a flexible tube and a plurality of liquid filters, wherein the plurality of liquid filters are positioned in predetermined distances from each other throughout the flexible tube.

Inventors: **Coccellato; Cannon (Sherman Oaks, CA)**

Applicant: **Coccellato; Cannon** (Sherman Oaks, CA)

Family ID: **96948666**

Assignee: **INTERGALACTIC SERVICES 24/7, LLC (Sherman Oaks, CA)**

Appl. No.: **18/600533**

Filed: **March 08, 2024**

Publication Classification

Int. Cl.: **B01D29/56** (20060101); **B01D29/05** (20060101); **B01D29/62** (20060101); **B01D35/30** (20060101); **B01D61/58** (20060101); **B01D65/02** (20060101); **C02F1/00** (20230101); **C02F1/44** (20230101); **C02F103/00** (20060101)

U.S. Cl.:

CPC **C02F1/44** (20130101); **B01D29/05** (20130101); **B01D29/56** (20130101); **B01D29/62** (20130101); **B01D35/306** (20130101); **B01D61/58** (20130101); **B01D65/02** (20130101); **C02F1/001** (20130101); B01D2317/02 (20130101); C02F2103/001 (20130101); C02F2201/001 (20130101); C02F2303/16 (20130101)

Field of Classification Search

CPC: B01D (29/05); B01D (29/56); B01D (29/62); B01D (35/306); B01D (36/04); B01D (61/58); B01D (65/02); B01D (2317/02); C02F (1/001); C02F (1/44); C02F (2103/001); C02F (2201/001); C02F (2303/16); E03B (3/02); E04D (2013/086)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
516968	12/1893	Gardner	114/349	B63C 9/06
3002870	12/1960	Belgrade et al.	N/A	N/A
5554277	12/1995	Rief et al.	N/A	N/A
6576123	12/2002	Priggemeyer et al.	N/A	N/A
8438788	12/2012	Bell et al.	N/A	N/A
8663465	12/2013	Harrison et al.	N/A	N/A
9243386	12/2015	Perlatti	N/A	N/A
9568211	12/2016	Wright et al.	N/A	N/A
2004/0002270	12/2003	Courtney	N/A	N/A
2010/0108617	12/2009	Schmidt et al.	N/A	N/A
2010/0244339	12/2009	Yoon et al.	N/A	N/A
2011/0005602	12/2010	Harrington	N/A	N/A
2011/0024341	12/2010	Lyon	N/A	N/A
2011/0100887	12/2010	Balistreri	N/A	N/A
2011/0211976	12/2010	Magoon	417/53	B01D 65/10
2011/0232766	12/2010	Bell	52/16	E04D 13/08
2020/0398204	12/2019	Vo	N/A	B01D 46/0005

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
210001674	12/2019	CN	N/A
110755923	12/2019	CN	N/A
215630336	12/2021	CN	N/A
1034828	12/1999	EP	B01D 29/118
10-0856883	12/2007	KR	N/A
10-1344852	12/2012	KR	N/A
WO-2016111610	12/2015	WO	B01D 61/18

OTHER PUBLICATIONS

EP1034828A1—EPO Machine Translation (Year: 2024). cited by examiner

Primary Examiner: Orme; Patrick

Attorney, Agent or Firm: Trojan Law Offices

Background/Summary

FIELD OF THE INVENTION

(1) The present invention relates to rain water collection systems, and in particular, to a rain water collection and filtration unit connected to a mobile water tank, wherein the water tank is capable of floating on water.

BACKGROUND

(2) Due to the present water crisis in the world, the use of rain water harvesting systems is of great importance. According to a UNICEF document, titled Water Scarcity, four billion people, almost two thirds of the world's population, experience severe water scarcity for at least one month each year. The document further provides that over two billion people live in countries where water supply is inadequate, and half of the world's population could be living in areas facing water scarcity by as early as 2025.

(3) Rain water harvesting systems are used worldwide. In such systems, rain water can be captured, filtered, and held for future use. The filtered water can be used as drinking water.

(4) Various documents presently exist that describe arrangements relating to rainwater collection. For example, Harrison discloses a filtration and dispensing system that is particularly suitable for underdeveloped parts of the world is provided for storing, filtering, and dispensing rainwater, surface water, or groundwater, wherein the system includes a bank of ceramic filters, a water retention container for holding water to be dispensed by the system, and at least one header tank to which water supplied from the water retention container is fed. (Harrison et al., U.S. Pat. No. 8,663,465 B2).

(5) It is an object of the present invention to provide a low cost and an easy to install rain water collection, filtration, and life boat system that can be used year round. The present invention serves an economical purpose during normal rainy seasons by allowing rainwater, which is usually absorbed into the ground or washed away, to be collected and filtered for human use and consumption. But the invention also serves a crucial lifesaving purpose in the event of a natural disaster, such that the buoyant water storage tank functions as a life raft with its own fresh water supply. The present rain water collection, filtration, and life boat system would be of great utility in parts of the world where water is scarce and/or seasonal flooding is an unfortunate part of life. In the event of a flood in rural India, for example, the elderly, young children, and/or pets who cannot swim can rest on the seating area of the life boat until the flood is over. Additional people can hang on to the safety ropes on the side of the storage tank/lifeboat to remain afloat without having to tread water indefinitely. All the while, fresh water, which is often in short supply after a natural disaster, would be readily available. Thus, the Water Kit has the potential to not only save money through harvesting rainwater for daily use, but to also save lives in the event of flooding following a natural disaster such as a hurricane, cyclone, typhoon, or tsunami.

SUMMARY

(6) The present invention provides a rain water collection, filtration, and life boat system, the system comprising a rain water collection surface having an outlet opening, a life boat having a water tank, and a connecting hose. The water tank has an inlet opening. The connecting hose connects the outlet opening to the inlet opening. The connecting hose is configured to allow a flow of collected rain water from the rain water collection surface to the water tank. The connecting hose comprises a flexible tube and a plurality of liquid filters. The plurality of liquid filters are positioned in predetermined distances from each other throughout the flexible tube.

(7) The plurality of liquid filters can be mesh screen filters or membrane filters. The plurality of liquid filters can be fitted into the interior of the connecting hose by a plurality of retention mechanisms. Each one of the plurality of retention mechanisms can be marked with a type indicator indicative of a type of a filter that can be fit into the interior of the connecting hose.

(8) Each one of the retention mechanisms comprises a curved clamp to hold one of the plurality of liquid filters in place, and the curved clamp is configured to lock/unlock by a latch. The life boat

can have a seating area.
(9) The invention further provides a flexible hose for rain water filtration, comprising a flexible tube defining a liquid passageway between a first end and a second end, wherein the first end of the flexible tube defines a first opening that is connectable to a source of a liquid. The second end of the flexible tube defines a second opening that is connectable to a water tank. A plurality of liquid filters can be attached to the interior of the flexible hose between the first and second ends to filter undesired matters in the liquid through the hose. The plurality of liquid filters are positioned in predetermined distances from each other throughout the flexible tube. The plurality of liquid filters can be mesh screen filters or membrane filters.
(10) The plurality of liquid filters are fitted into the interior of the flexible hose with a plurality of retention mechanisms. Each one of the retention mechanisms comprises a curved clamp, wherein the curved clamp holds one of the plurality of filters in place, and the curved clamp is configured to lock/unlock by a latch. Each one of the plurality of retention mechanisms can be marked with a type indicator indicative of a type of a filter that can be fit into the interior of the flexible hose. The flexible hose may further comprise a pair of male tracks positioned on opposite sides of the flexible tube. The plurality of liquid filters can be replaced by unlocking and locking the curved clamp.
(11) The invention further provides a method of rain water collection and filtration, the method comprising the steps of collecting rain water with a collection surface, inserting a plurality of liquid filters into predetermined locations in a flexible connecting hose, and allowing the collected rain water to flow through the flexible connecting hose from the collection surface to a floatable water tank. The method may further comprise the steps of the determining an efficiency of the plurality of liquid filters, and replacing at least one of the plurality of the liquid filters based on the determined efficiency. The method may further comprise the steps of determining an efficiency of the plurality of liquid filters, removing at least one of the plurality of the liquid filters based on the determined efficiency, cleaning the at least one of the plurality of the liquid filters, inserting the at least one of the plurality of the liquid filters into the flexible connecting hose. In various implementations of the invention, the order of the steps of the method can be changed, and some steps can be performed simultaneously.

Description

BRIEF DESCRIPTION OF THE FIGURES
(1) FIG. **1** illustrates a perspective view of a rain water collection, filtration, and life boat system.
(2) FIG. **2** illustrates a perspective view of a rain catcher and a stand for the rain catcher.
(3) FIG. **3** illustrates a perspective view of a connecting hose.
(4) FIG. **4** illustrates a perspective view of a portion of the connecting hose, and a protective female hose.
(5) FIG. **5** illustrates a perspective view of a retention mechanism of the connecting hose in open position.
(6) FIG. **6** illustrates a perspective view of the retention mechanism of the connecting hose in locked position.
(7) FIG. **7** illustrates a cross section view of the connecting hose wherein a filter can be inserted/removed via the retention mechanism.
(8) FIG. **8** illustrates a cross section view of the connecting hose wherein a filter is positioned within the connecting hose.
(9) FIG. **9** illustrates a perspective view of a life boat having a water tank and a seating area.
DESCRIPTION
(10) The present invention is described more fully hereinafter, but not all embodiments are shown. While the invention has been described with reference to exemplary embodiments, it will be

understood by those skilled in the art that various changes may be made, and equivalents may be substituted for elements thereof without departing from the scope of the disclosure. In addition, many modifications may be made to adapt a particular structure or material to the teachings of the disclosure without departing from the essential scope thereof.

(11) The drawings accompanying the application are for illustrative purposes only. They are not intended to limit the embodiments of the present application. Additionally, the drawings are not drawn to scale. Common elements between different figures may retain the same numerical designation.

(12) Referring to FIG. **1**, the figure illustrates a perspective view of a rain water collection, filtration, and life boat system, comprising a rain catcher **10**, a stand **20**, connecting hose(s) (**30**, **60**), and a life boat **70**. The rain catcher **10** can be positioned on the stand **20**. The rain catcher **10** is adapted to collect the rain water through its top opening area. The collected rain water can be filtered through a plurality of filters positioned on the top and inside the rain catcher **10**. The collected rain water can be further filtered by additional filters that are positioned inside the connecting hose(s). The connecting hose(s) (**30**, **60**) connect outlet openings of the rain catcher **10** to inlet openings of the water tank **71** of the life boat **70**. The collected rain flows through connecting hose(s) (**30**, **60**) to the water tank **71**. In its grounded position, the life boat **70** may be positioned on a plurality of legs **76**. In its floating position, the connecting hose(s) can be adapted to be easily detached from the water tank **71**. The connecting hose(s) can be connected to the water tank of a single life boat as shown in FIG. **1** or can be connected to a plurality of life boats.

(13) Referring to FIG. **2**, the figure illustrates a perspective view of the rain catcher **10** and its stand **20**. The rain catcher comprises side walls (**11***a*, **11***b*, **11***e*, **11***d*), and a bottom surface **11***c*. A top opening **17** is formed on the top of the rain catcher **10**, wherein the top opening **17** is formed between the top of the side walls (**11***a*, **11***b*, **11***e*, **11***d*). The rain catcher **10** may further comprise a pair of lips (**12***a*, **12***b*) for a lid (not shown) to slide over when the rain catcher is not in use. The rain catcher **10** catches the rain through its top opening **17**. The rain catcher may further have a plurality of screen/mesh filters such as the top exterior filter **14** and the bottom interior filter **13**, wherein the bottom interior filter is inserted through the top opening **17** to be placed inside the rain catcher **10**, and the top exterior filter is positioned on the top or in proximity of the top of the rain catcher **10**. The plurality of screen/mesh filters are used to separate fine particles and sand out of the collected water. The mesh/screen filters of the rain catcher **10** can be made of flexible or rigid materials. For example, the filter can be a non-rust, thick mesh filter that is placed on top of the rain catcher. The rain catcher **10** may further comprise a pair of spigots (**15***a*, **15***b*) with flow control handles. In an embodiment of the invention, the rain catcher **10** is made from fiber-reinforced plastic (FRP) and polyethylene. The rain catcher **10** can be made in different sizes. In an implementation of the invention, the opening **17** can be a 4 ft by 8 ft opening.

(14) Still referring to FIG. **2**, the rain catcher **10** further comprises one or more outlet openings such as **16***a*, **16***b*, **16***c*, and **16***d*. The outlet openings can be female adapters to receive one end of the connecting hose(s) (not shown).

(15) As shown in FIGS. **1-2**, the rain catcher **10** can be positioned on a stand **20**. The stand **20** may comprise side walls (**21***a*, **21***b*) and a bottom surface **21***c*. The shape of the stand **20** conforms to the shape of the rain catcher **10** so the rain catcher **10** can fit inside the stand **20**. The stand **20** further comprises holding legs **27**. The holding legs **27** may have a height adjustment mechanism such as an affixing pin **30**, a hollow beam **28** with first set of apertures, and a slide-in beam **29** with second set of apertures, wherein the affixing pin **30** can be inserted through one of the first set of apertures and one of the second set of apertures to adjust the height of one of the holding legs **27**. A plurality of openings/cut-outs such as **26***a*, **26***b*, **26***c*, and **26***d* can be created at the bottom of the stand to allow for insertion of one or more outlet openings (such as **16***a*, **16***b*, **16***c*, and **16***d*) of the rain catcher **10** when is placed on the stand **20**.

(16) Referring to FIG. **3**, the figure illustrates a perspective view of the connecting hose **30**. The

connecting hose **30** comprises a flexible tube **31** defining a liquid passageway between a first end **32***a* and a second end **32***b*. The first end **32***a* of the flexible is connectable to a source of a liquid such as the rain catcher **10** as shown in FIG. **1**. The second end of the flexible tube **31** is connectable to a water tank such as the one (**71**) shown in FIG. **1**.

(17) A plurality of retention mechanisms (such as **50***a*, **50***b*, and **50***c*) are coupled to the exterior of the connecting hose **30** in predetermined distances from each other. Each retention mechanism is capable of holding a liquid filter in place inside the connecting hose **30**, such that the liquid flowing through the connecting hose is filtered multiple times by passing through multiple liquid filters. Each retention mechanism is adapted to receive a liquid filter such as a screen/mesh/membrane filter as shown in FIGS. **5** and **8** (filter **39**). Therefore, a plurality of liquid filters (not shown) can be attached to the interior of the connecting hose **30** in predetermined distances from each other between the first end **32***a* and the second end **32***b* to filter undesired matters in the liquid passing through the connecting hose **30**. The plurality of liquid filters can be membrane filters with various micron sizes, wherein the closest membrane filter to the first end **32***a* has the highest micron size, and the micro size of the membrane filters can gradually decrease throughout the connecting hose to a 1 micron size for the closest filter to the second end **32***b*. Additionally, a liquid filter **35** can be coupled to the first end **32***a*, and another liquid filter (not shown) can be coupled to the second end **32***b*. The connecting hose **30** may further comprise a plurality of elongated male tracks (such as **33***a* and **33***b*). The connecting hose **30** can either function by itself, or can be further covered by a protective female hose **40** as shown in FIG. **4**.

(18) In an embodiment of the invention, the outside of the connecting hose **30** may be made from a coating material coating such as fiber-reinforced plastic material (FRP), and the interior of the connecting hose can be made from polyethylene resin, green or black, to reduce algae growth.

(19) Referring to FIG. **4**, the figure illustrates a perspective view of a portion of the connecting hose **30**, and a protective female hose **40**. As shown in the figure, the connecting hose **30** may further comprise a plurality of filter holders (such as **36***a*, **36***b*, **36***c*), wherein each one of the filter holders is adapted to receive a liquid filter. The plurality of retention mechanisms (such as **50***a*, **50***b*, and **50***c*) are coupled to the plurality of filter holders (such as **36***a*, **36***b*, **36***c*) respectively, wherein a filter (not shown) such as a screen/mesh filter can be placed within the each internal frame, and can be held in place by the corresponding retention mechanism.

(20) Still referring to FIG. **4**, the connecting hose **30** can slide inside a protective female hose **40**, wherein the protective female hose **40** provides further strength or protection for the internal connecting hose. The protective female hose may have an elongated body **41**, and a plurality of female tracks (such as a pair of female tracks **46***a* and **46***b*), where the opening **42** and interior of the protective female hose **40** can be adapted to receive the connecting hose **30**. The female tracks (**46***a*, **46***b*) can be adapted to receive the male tracks of the connecting hose **30**. The protective female hose **40** may further have a plurality of opening (such as **43**) with opening covers (such as **44**). The opening cover **44** can be a sliding cover that can be moved in opposite directions **45** to provide access to the retention mechanism of the connecting hose that is positioned under the opening cover **44** on the connecting hose.

(21) Referring to FIGS. **5-6**, the figures illustrate a perspective view of a retention mechanism **50***a* of the connecting hose **30** in unlocked position (FIG. **5**) and locked position (FIG. **6**). The retention mechanism (**50***a* in FIG. **6**) may comprise a clamp such as the curved clamp **51** and a watertight latch **53**, wherein the curved clamp **51** and the latch **53** can be engaged/disengaged to provide a locking/unlocking mechanism for the retention mechanism. As shown in FIG. **5**, the connecting hose comprises a filter insertion opening **37** which can be covered or uncovered by the clamp **51**. When the retention mechanism is in unlocked position (FIG. **5**), a liquid filter **39** can be inserted into the connecting hose **30** through the filter insertion opening **37**, or can be removed therefrom. When the retention mechanism is in locked position (FIG. **6**), the clamp **51** can hold a liquid filter in place inside the connecting hose **30**. In a preferred embodiment of the invention, the retention

mechanism is a watertight retention mechanism.
(22) Still referring to FIGS. **5-6**, in an implementation of the invention, a type indicator **52** can be marked in proximity of or on each retention mechanism (such as **50***a* as shown in FIG. **6**) to indicate the type of the filter that can be held in place by the marked retention mechanism. For example, the exterior of the curved clamp **51** can be marked with a specific color or a code (**52**) indicative of the type of filter that can be used with the marked retention mechanism. Similarly, a type indicator can be marked on each liquid filter to indicate the type of the filter. The liquid filter (such as **39**) can be a color-coated membrane wherein the color determines the micron size of the membrane filter. Various colors can be used to indicate various micron sizes. The liquid filter **39** can be a disk-shaped filter that is adapted to fit inside the connecting hose **30**. The liquid filter **39** can be a framed membrane filter, wherein the filter is being held by a frame **38**.
(23) Referring to FIG. **7**, the figure illustrates a cross section view of the connecting hose **30** when the retention mechanism is in its unlocked/open position. When the retention mechanism is in unlocked position, the filter **39** can be inserted into or removed from the connecting hose **30** (see arrows **56**) through the filter insertion opening **37**. The connecting hose **30** may have a plurality of filter holders (such as **36***a*, **36***b*, **36***c* as shown in FIG. **4**) for holding liquid filters. A representative filter holder **36***a* (as shown in FIG. **7**) is adapted to receive the frame **38** of the liquid filter **39**. As shown in FIG. **8**, when the liquid filter **39** is positioned within the connecting hose **30**, the retention mechanism (clamp **51** and the latch **53**) can be locked (as shown by arrows **57** and **58**) to hold the liquid filter in place inside the connecting hose.
(24) Referring to FIG. **9**, the figure illustrates a perspective view of a life boat **70** having a water tank **71**, a seating area **74**, backrest area **79**, step **80**, safety paddles **77**, safety handles **75**, seat handles **81**, and backrest handles **82**.
(25) The water tank **71** may have a plurality of inlet openings **72** and a plurality of water dispensers **73**. The inlet openings **72** are adapted to connect to the connecting hose (not shown). The inlet openings **72** may have female tracks to receive the male tracks of a connecting hose. The inlet opening can be a quick release valve capable of being easily detached from the connecting hose. As illustrated in FIG. **1**, the connecting hose(s) (**30**, **60**) connect the water tank **71** of the life boat **70** to the outlet openings of the rain catcher **10**, to allow the collected rain water flow into the water tank **71**. The water tank **71** may have additional filters (not shown) to enhance the quality of the drinking water.
(26) As shown in an embodiment of the invention in FIG. **9**, the seating area **74** can be a 360° seating area with the 360° backrest area **79**. The safety paddles **77** can be attached to all sides of the life boat. The life boat **70** can further have a 360° step **80** at the base of the life boat, and a plurality of handles such as safety handles **75**, seat handles **81**, and backrest handles **82**, wherein a person can grab onto the handles and climb up the step to reach the seating area. Handles can be cut-out handles such as the cut-out backrest handles **82**.
(27) The life boat can be made from fiber-reinforced plastic (FRP). In an implementation of the invention, a percentage or some parts of the life boat can be made from low-density polyethylene (LDPE), high-density polyethylene (HDPE), and/or Polypropylene (PP). In another implementation of the invention, a percentage or some parts of the life boat can be made from recycled low-density polyethylene (LDPE), recycled high-density polyethylene (HDPE), and/or recycled Polypropylene (PP) to help recycle the plastic on the planet.
(28) The invention serves a crucial lifesaving purpose in the event of a natural disaster, such that the life boat **70** with its own water supply/tank **71**. The present rain water collection, filtration, and life boat system would be of great utility in parts of the world where water is scarce and/or seasonal flooding is an unfortunate part of life. In the event of a flood, the people (**78**) in need can sit on the seating area **74** of the lifeboat **70** until the flood is over. Additional people can hang on to the handles on the side of the lifeboat **70** (such as the safety handles **75**) to remain afloat. All the while, fresh water, which is often in short supply after a natural disaster, would be readily available

through water dispensers **73** which are connected to the water tank **71**. Thus, the invention has the potential to not only save money through harvesting rainwater for daily use, but to also save lives in the event of flooding following a natural disaster such as a hurricane, cyclone, typhoon, or tsunami.

(29) The foregoing descriptions of embodiments of the present invention have been presented only for purposes of illustration and description. They are not intended to be exhaustive or to limit the present invention to the forms disclosed. Accordingly, many modifications and variations will be apparent to practitioners skilled in the art. Additionally, the above disclosure is not intended to limit the present invention. The scope of the present invention is defined by the appended claim.

Claims

1. A flexible hose for rain water filtration, comprising: a flexible tube defining a liquid passageway between a first end and a second end; the first end of the flexible tube defining a first opening connectable to a source of a liquid; the second end of the flexible tube defining a second opening connectable to a water tank; a plurality of filter insertion openings positioned in predetermined distances from each other along the flexible tube, each comprising a passageway into and out of the side of the flexible tube; and a plurality of liquid filters, each configured to be inserted cross-sectionally from outside the flexible tube through one of the plurality of filter insertion openings and into the flexible tube, thereby positioning each of the plurality of liquid filters at a predetermined distance from each other along the length of the flexible tube; and a plurality of retention mechanisms configured to individually cover each filter insertion opening, each retention mechanism comprising a clamp and a watertight latch, each retention mechanism configured to cover one of the plurality of filter insertion openings, such that each of the plurality of retention mechanisms can secure a filter in the flexible tube and create a watertight seal with the flexible tube.
2. The flexible hose of claim 1, further comprising a first-end liquid filter and a second-end liquid filter, the first-end liquid filter is coupled to the first end of the flexible tube, and the second-end liquid filter is coupled to the second end of the flexible tube.
3. The flexible hose of claim 1, wherein the plurality of liquid filters comprises mesh screens.
4. The flexible hose of claim 1, wherein the plurality of liquid filters comprises membrane filters with various micron sizes, wherein one of the membrane filters that is closest to the first end has the highest micron size.
5. The flexible hose of claim 1, wherein each of the clamps comprises a curved clamp holding one of the plurality of liquid filters in place, and wherein each curved clamp is configured to be releasably lockable by a corresponding watertight latch.
6. The flexible hose of claim 5, wherein each of the plurality of liquid filters is configured to be replaced by unlocking and locking the curved clamp.
7. The flexible hose of claim 1, wherein each one of the filter insertion openings is slot-shaped.
8. The flexible hose of claim 1, wherein the flexible hose further comprises a pair of male tracks positioned on opposite sides of the flexible tube.
9. A rain water collection, filtration, and life boat system, the system comprising: a rain water catcher having side walls, a bottom surface, an open top and at least one outlet opening; an adjustable height stand for holding the rain water catcher; a life boat having a water tank, the water tank having an inlet opening; a connecting hose, the connecting hose connecting the outlet opening to the inlet opening, the connecting hose configured to allow a flow of collected rain water from the rain water catcher to the water tank; the connecting hose comprising a flexible tube with a plurality of filter insertion openings, each comprising a passageway into and out of the side of the flexible tube, a plurality of liquid filters, each configured to be inserted cross-sectionally from outside the flexible tube through one of the filter insertion openings into the flexible tube, thereby positioning

each of the plurality of liquid filters at a predetermined distance along the length of the flexible tube; and a plurality of watertight retention mechanisms, each retention mechanism configured to cover one of the plurality of filter insertion openings, such that each of the plurality of retention mechanisms can secure a filter in place in the flexible tube.

10. The rain water collection, filtration, and life boat system of claim 9, wherein the plurality of liquid filters each comprise a mesh screen.

11. The rain water collection, filtration, and life boat system of claim 9, wherein the plurality of liquid filters comprises membrane filters.

12. The flexible hose of claim 9, wherein each of the plurality of retention mechanisms is marked with a type indicator indicative of a type of filter that fits into the interior of the connecting hose.

13. The rain water collection, filtration, and life boat system of claim 9, wherein each one of the retention mechanisms comprises a curved clamp to hold one of the plurality of liquid filters in place, and the curved clamp is configured to be releasably lockable with a latch.

14. The rain water collection, filtration, and life boat system of claim 9, the life boat further having a seating area.